

Chapter 1: Introduction to Embedded Systems

- Embedded System Overview
- Classification of Embedded Systems
- Hardware and Software in a System
- Purpose and Application of Embedded Systems

200

1.1 Embedded System Overview

A. Introduction

An embedded system is a combination of computer hardware and software and may have additional mechanical or other parts designed to perform a specific function. Embedded system is a kind of computer system that performs a dedicated function and/or is intended for use with a specific embedded software application. Embedded systems are devices which are used to control, monitor or assist the operation of an equipment, machinery or plant. An embedded system may be defined as a computer system designed for specific control functions within a larger system, often with real time computing constraints. It may be embedded as a part of a complete device often including hardware and mechanical parts.

B. Characteristics

Single functioned: As embedded systems are designed for specific control functions, it usually executes a specific program to carry out the specific function repeatedly. In some cases, exceptions may occur but in general all embedded systems are supposed to carry out single specified function. One case is where an embedded system program is updated with a newer version. A second case is when several programs are swapped in and out of a system due to some constraint.

Tightly constrained: Feasibility and utility of the embedded system are measured in terms of cost, size, performance, power and other parameters. These all parameters are referred as design metric.

All computing systems are constraint with design metric but it is more tightly constraint in embedded systems. Embedded systems in general must be economic, small possible size, fast enough to process data in real time and must consume minimum power to extend battery life or prevent the necessity of a cooling fan.

Reactive and real time: In general embedded systems must continually react to changes in the system's environment and must compute certain results in real time without delay. A delay in computation and slow response may result a failure in the operation of the system.

C. Components

An embedded system has mainly three components hardware, application software and real time operating system (small scale embedded system may not require RTOS).

Hardware: It represents the physical component of the system which interacts with each other to perform the specific task. Processor, RAM, ROM, ADC, DAC, Timers, Ports etc are some of the hardware components of the embedded system.

Application Software: The application software may perform concurrently the series of tasks or multiple tasks. Generally they are written in Assembly, C, C++, Java etc.

RTOS: It supervises the application software and provides a mechanism to let the processor run a process as per scheduling and do the context-switch between various tasks. RTOS defines the way the system works and sets the rules during the execution of the application software. Win CE, VxWorks, Embedded Linux etc.

D. Design Metrics

A design metric is a measurable feature of a system's implementation. Some of the commonly used metrics include:

- **NRE cost** (nonrecurring engineering cost): It represents the ^{consisting of money} monetary cost for designing the system. Large number of units can be produced without any additional design cost. Since the cost doesn't occur more than once for a particular system, it is termed as nonrecurring.
- **Unit cost:** It is the monetary cost of manufacturing each unit of the system excluding NRE cost.
- **Size:** It is the physical space required by the system. For software It is measured in terms of bytes and for hardware it is measured in terms of no of gates or transistors.
- **Performance:** It represents the execution time of the system.
- **Power:** The amount of power consumed by the system, which may determine the lifetime of a battery, or the cooling requirements of the IC.
- **Flexibility:** The ability to change the functionality of the system without incurring heavy NRE cost.
- **Time to prototype:** The time needed to build a working version of the system, which may be bigger or more expensive than the final system implementation. It can be used to verify the system's usefulness and correctness and to refine the system's functionality.
- **Time to market:** The time required to develop a system to the point that it can be released and sold to customers.
- **Maintainability:** The ability to modify the system after its initial release.

- **Correctness:** We can check the functionality throughout the process of designing the system and we can insert test circuitry to check that manufacturing was correct.
- **Safety:** The system is supposed to cause no harm.

The Time to Market Design Metric

Introduction of an embedded system to the marketplace significantly affects the overall system profitability. The market window, period during which the product have highest sales, for products is getting shorter, so a short delay on introduction of product to the marketplace can render huge loss. Using a simplified model of revenue as shown in the figure below, we will deduce the loss of revenue that can occur due to delayed entry of a product in the market.

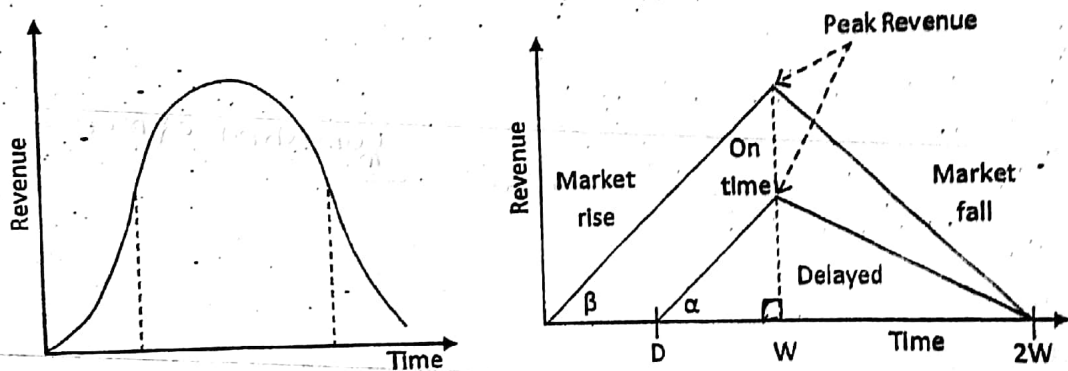


Figure 1.1: Market window and simplified revenue model for loss calculation for delayed entry

This model assumes the peak of the market occurs at the halfway point, denoted as W , of the product life. The peak is same for delayed entry. The revenue for an on-time market entry is the area of the triangle labeled On-time, and for delayed entry is the area of triangle labeled Delayed. The difference between the areas of two triangles gives the revenue loss for a delayed entry.

$$\text{Revenue Loss} = ((\text{On time} - \text{Delayed}) / \text{On time}) * 100$$

$$\begin{aligned} \text{Area of On time triangle} &= \frac{1}{2} * \text{base} * \text{height} \\ &= \frac{1}{2} * 2 * W * W * \tan \beta \quad (\text{Assuming, market rise angle is } \beta) \\ &= W^2 \tan \beta \end{aligned}$$

$$\text{Area of Delayed entry triangle} = \frac{1}{2} * (2W - D) * (W - D) * \tan \alpha$$

Assuming $\beta = \alpha$, and on solving we get,

$$\text{Revenue Loss} = (D(3W - D) / 2W^2) * 100\%$$

The NRE and Unit Cost Design Metrics

The NRE cost is the one time monetary cost of designing the system, whereas the unit cost represents the monetary cost of manufacturing each copy of the system, excluding NRE cost.

$$\text{Total Cost} = \text{NRE cost} + \text{unit cost} * \# \text{ of units}$$

$$\text{Per-product Cost} = \text{total cost} / \# \text{ of units}$$

$$= \text{NRE cost} / \# \text{ of units} + \text{unit cost}$$

The larger the volume, the lower the per-product cost, since the NRE cost can be distributed over more products. The per-product cost of the product approaches the unit cost for very large volume.

For Example, let us consider products using three different technologies; Technology A with NRE Cost of \$2000 and unit cost of \$100, B with NRE cost of \$30000 and unit cost \$30, and C with NRE cost of \$100000 and unit cost \$2

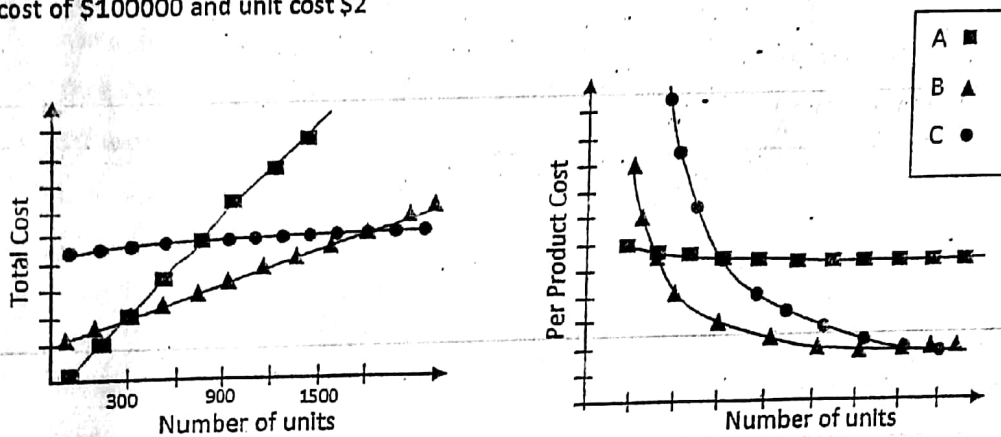


Figure 1.2: Plot of total and per product cost as a function of volume for products A, B and C

The plot on the left, total cost versus volume, shows that line of technology A and B intersect at 300 which implies technology A yields the lowest total cost for low volumes, less than 300 units. Technology B Yields lowest total cost for volume between 300 and 1800, since the line of technology B and C meets at 1800. Furthermore, technology C yields the lowest cost for volumes above 1800. The plot on the right, per-product cost versus volume, illustrates how larger volume will amortize NRE costs resulting in lower per-product costs. For example, for technology C with a volume of 50000, the per-product cost is \$4 but if we consider a volume of 200000, the per-product cost will reduce to \$2.5.

The Performance Design Metric

Performance of a system is a measure of duration the system takes to execute our desired tasks. Though the performance of a system is governed by clock frequency or Instructions per second, the main measures of performance are:

Latency or response time: the time between the start of the task's execution and the end.

Throughput: the number of tasks that can be processed per unit time.

Speedup is a common method of comparing the performance of two systems. The speedup of system A over system B is determined by:

Speedup of A over B = performance of A/performance of B

E. Example of an Embedded System – A Digital Camera

A digital camera can be taken as embedded system as it performs only a single function of capturing image. It is tightly constrained as it is affordable, portable, and consumes less power. And as it is fast enough to process numeral images in milliseconds, it exhibits real time feature. But however, a simple digital camera may not possess high degree of reactive attribute. On the contrary, few contemporary digital cameras are capable of detecting human expressions.

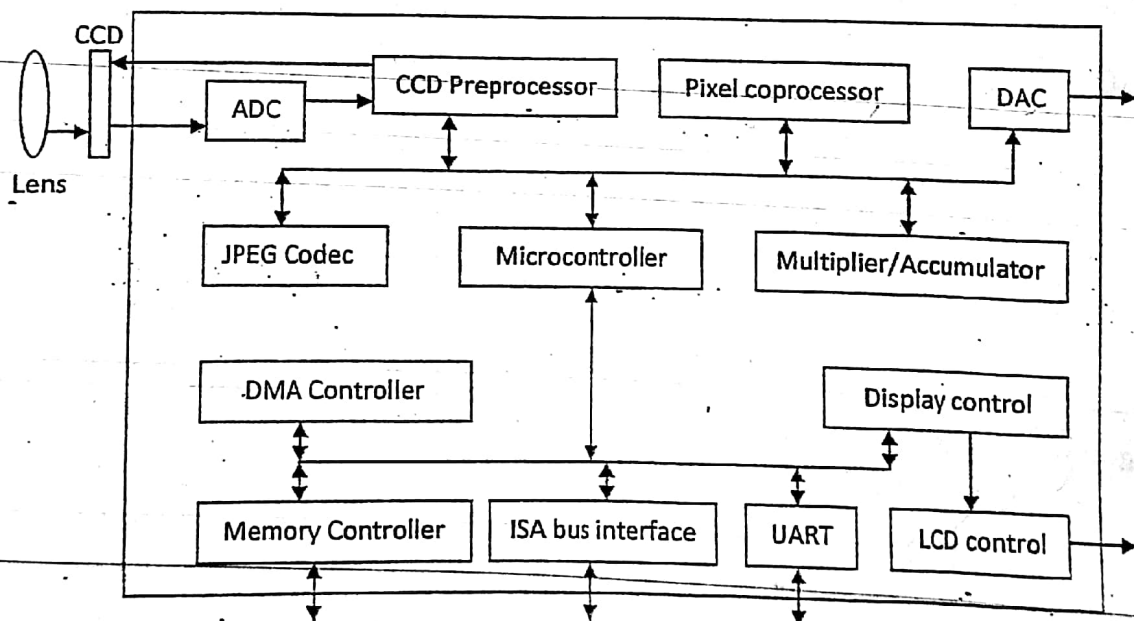


Figure 1.3: Block diagram of a typical digital camera

The different components of the digital camera chip are listed below:

- **Charge - Coupled Device (CCD):** It contains an array of light sensitive photocells that capture an image. A CCD is an integrated circuit etched onto a silicon surface forming light sensitive elements called pixels. Photons incident on this surface generate charge that can be read by electronics and turned into a digital copy of the light patterns falling on the device. Photons striking a silicon surface create free electrons through photoelectric effect. And the electrons are gathered in the place where they are generated and count them in some manner to create an image. It is accomplished by positively biasing discrete areas to attract electrons generated while the photons come onto the surface.
- **Analog to digital converter:** It converts analog images to digital images. The main job is to classify the voltages of the pixels into levels of brightness and to assign each level to a binary number, consisting of zeroes and ones. Most digital cameras use at least an 8-bit ADC, which allows for up to 256 values for the brightness of a single pixel.
- **Digital to analog converter:** It converts digital images to analog images.
- **CCD preprocessor:** it provides commands to the CCD to read the image. Some other functions of preprocessor are over scan correction and trimming, bias removal, dark current removal, and flat fielding.
- **JPEG (Joint Photographic Expert Group) Codec:** It compresses and decompresses an image using the JPEG compression standard which enables compact storage of images in the limited memory of the camera.
- **Pixel coprocessor:** It helps in rapidly displaying images. A pixel coprocessor is required in digital cameras for displaying images directly or after operations such as rotate right, rotates left, rotate up, rotate down, shift to next, shift to previous.
- **Memory Controller:** It controls the access to a memory chip. It is a digital circuit which manages the flow of data going to and from the main memory.
- **DMA (Direct Memory Access) Controller:** It enables direct memory access by other devices while the microcontroller is performing other functions.
- **UART (Universal Asynchronous Receiver and Transmitter):** It enables communication with a PC's serial port for uploading video frames. It is the microchip with programming that controls a computer's interface to its attached serial devices.
- **ISA (Industry Standard Architecture) bus interface:** It enables faster connection with a PC's ISA bus. Most PC's have an ISA slot on the main board that accepts either an 8 bit or a 16 bit ISA printed circuit card.

- **LCD Control:** It controls the display of images on the camera's liquid crystal device.
- **Display Control:** It also controls the display of images on the camera's liquid crystal device.
- **Multiplier/Accumulator:** It assists with certain digital signal processing.
- **Microcontroller:** It is the processor that controls the activities of other circuits within the system.

1.2 Classification of Embedded Systems

Classification based on Generation

A. First Generation

Embedded systems were designed using 8 bit microprocessors or 4 bit microcontrollers. The hardware circuits were simple and the firmware was developed using assembly code. Motor controller using 8085 can be an example of first generation embedded system.

B. Second Generation

In second generation, the systems were built using 16 bit microprocessors and 8/16 bit microcontrollers. More complex and powerful instructions were available for the designer to work with. Some systems involved embedded operating systems for their operation. Data Acquisition Systems can be an example of second generation embedded systems.

C. Third Generation

The systems were designed with more advanced processor technology in the form of 32 bit processors and 16 bit microcontrollers. Along with complex and powerful instruction sets, instruction pipelining was introduced for better performance. Dedicated embedded real time operating system implementation was another important feature in this generation. Also, the concept of application specific processors like Digital Signal Processors (DSP) and Application Specific Integrated Circuits (ASIC) came into existence.

D. Fourth Generation

Fourth generation was marked with the advent of System on Chips (SoC), reconfigurable processors and multicore processors. These embedded systems used high performance real time embedded operating systems for its operation.

Classification Based on Complexity and Performance

A. Small Scale Embedded Systems

These systems are designed with a single 8 or 16 bit microcontroller (8051 family, PIC16F8X, Hitachi H8). They have little hardware and software complexities and involve board level design. They may be battery operated. While developing embedded software for these system, an editor, assembler and cross assembler specific to the microcontroller or processor are used as the main programming tool. Usually C language is used for developing these systems. To develop such systems, the size requirement of the software must not exceed the available memory while the hardware design must be done in such a way that the power dissipation must be limited when the system is running continuously. Automatic vending machine, stepper motor controller for a robotics system etc can be the examples of small scale embedded systems.

B. Medium Scale Embedded Systems

These systems are designed with a single or few 16 or 32 bit microcontrollers (8051MX, PIC16F876) or DSPs or Reduced Instruction Set Computers (RISCs). It may also employ the readily available single purpose processors and IPs for various functions, for example: bus interfacing, encryption, deciphering and so on. These systems have both hardware and software complexities. For software design, the programming tools used is RTOS, Source code engineering tool, Simulator, Debugger and Integrated Development Environment (IDE). Software tools also provide the solutions to the hardware complexities. Some of the examples of medium scale embedded systems are Computer networking systems, signal tracking system etc.

C. Sophisticated Embedded Systems or Large Scale Embedded Systems

These systems have enormous hardware and software complexities and may need scalable processors or configurable processors and programmable logic arrays. They are used for cutting edge applications that need hardware and software co-design and integration in the final system. They are constrained by the processing speeds available in their hardware units. Certain software functions are implemented in the hardware to obtain additional speed by saving time. Some of the functions of the hardware resources in the system are also implemented by the software. These systems generally implement high performance real time operating system. Development tools for these systems may not be readily available at a reasonable cost or may not be available at all. In some cases, a compiler or retargetable compiler might have to be developed for these. (A retargetable compiler is one that configures according to the given target configuration in a system). Embedded System for wireless LAN & for convergent technology devices is one of the sophisticated embedded systems.

1.3 Hardware and Software in a system

In an embedded system, hardware of the system represents the single purpose processor whereas the software represents the general purpose Processor.

A. Single Purpose Processor

Single Purpose processor is a digital circuit designed to execute exactly one program. It does not require a program memory since the program to run on the processor is fixed (only one) and it can be built into the digital circuit. The datapath contains only the essential components for the specified task. JPEG codec, Display controller, DMA controller etc can be taken as the examples of single purpose processor.

Design metric benefits of single purpose processor

- Performance may be fast, size and power may be small
- Unit cost may be low for large quantities

Design metric drawbacks of single purpose processor

- Design time and NRE costs may be high, flexibility low, unit cost high for small quantities
- Performance may not match general purpose processors for some applications.

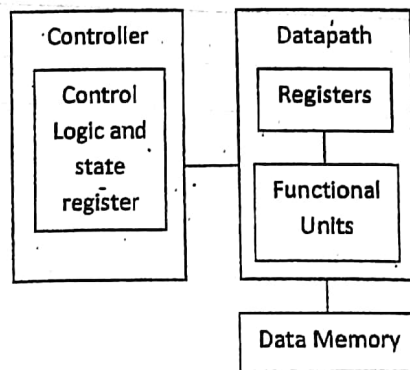


Figure 1.4: Block Diagram of Single Purpose Processor

B. General Purpose Processor

The required functionality is carried out by programming the processor's memory. In this context, a programmable device is built that is suitable for a variety of applications. Microprocessors are the examples of general purpose processor. Such processors include:

- **Program memory:** The program cannot be built into the digital circuit in general purpose processors since the program likely to run on the processor will be unknown.
- **General Datapath:** The datapath must be general enough to handle a variety of computations, so such datapath typically has a large register file and one or more general purpose arithmetic logic units (ALUs).

Design metric benefits of general purpose processor

- Time to market and NRE costs are low because designer must only write a program but does not have to deal with any digital design.
- Flexibility is high because changing functionality requires changing only the program.
- Unit cost may be low in small quantities.
- Performance may be fast for computation intensive applications.

Design metric demerits of general purpose processors

- Unit cost may be relatively high for large quantities, since in large quantities we could design our own processor and amortize NRE costs.
- Size and power may be large due to unnecessary processor hardware.
- Performance may be slow for certain applications.

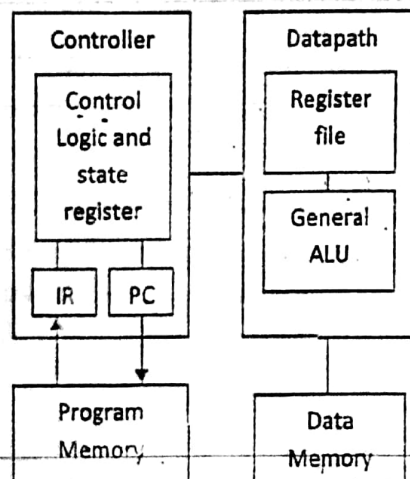


Figure 1.5: Block Diagram of General Purpose Processor

C. Application Specific Processors

Application specific processors are programmable processors optimized for a particular class of applications. It generally includes program memory, optimized datapath and special functional units. These processors provide optimum level of performance maintaining appropriate size and power consumption. Microcontrollers for controlling application and digital signal processors (DSPs) for huge data processing application are examples of application specific processors.

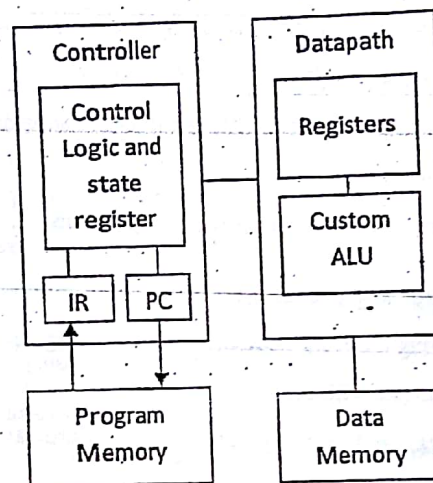


Figure 1.6: Block diagram of Application Specific Processors

1.4 Purpose and Application of Embedded Systems

Purpose of Embedded Systems

A. Data collection

In embedded systems, the data is collected from other external devices for storage, analysis, manipulation or transmission. Data may be in analog or digital form. Systems working with digital data require analog to digital converters if the collected data is in analog form. The collected data can be used for meaningful purpose based on the functionality of the embedded system. For instance, a digital camera collects data, stores it and finally provides graphical representation of data in the form of captured image.

B. Data communication

An embedded system is required to connect two or more devices which may be at close vicinity or at remote location. The communication between devices can be done via wired line medium or wireless medium. Embedded systems are incorporated with different wireless modules or wire-line

modules for communication purpose. Bluetooth, ZigBee, Wi-Fi, and GSM are few wireless modules. And for wire-line purpose, an embedded system may have RS-232, SPI, I2C, USB and other serial and parallel protocols. Some embedded systems like network hubs, routers, etc act as mediators in data communication and provide various features including data security.

C. Data Processing

The collected data in embedded system is subjected to some sort of processing for which embedded systems are attributed with data processing modules. Speech coder, audio video codec, etc can be few examples of data processing unit. Data processing includes the manipulation of data for appropriate purpose.

D. Monitoring

Many embedded systems are incorporated with sensors to check the state of the different parameters. The parameters can be current, voltage, temperature, humidity, etc which are continuously monitored and appropriate processing or controlling of devices is done. However, the value of the parameters cannot be controlled by the system itself. The values of parameters are used for some controlling purpose or for some graphical representation purpose or simply stored for further analysis and processing.

E. Control

For control purpose, actuators along with sensors are present in the embedded systems. The sensor connected in input port detects the change in the desired parameter and the actuators at output ports are controlled accordingly to implement the desired functionality. Electric Motors are examples of actuators. In an object avoiding robot, ultrasonic sensor senses the presence of certain kind of object and the motor is rotated accordingly to avoid the collision.

F. Application specific user interface

To provide the better user interface based on application has been one of the concerns of contemporary embedded systems. Keypads, simple LCD modules, speakers, etc are basic and common interface for users. However, sensitive touch pad along with high definition display has been the sophisticated interface implemented in current scenario.

Applications of Embedded Systems

1. Household appliances: Microwave ovens, Television, DVD players and recorders.
2. Consumer electronics: cell phones, cameras, video games

3. Office automation: fax machines, printers, scanners
4. Business equipment: alarm systems, card readers,
5. Automobiles: engine controller, fuel injection, antilock brakes.
6. Networking: Modem, Network cards, Network switches and routers
7. Medical equipments
8. Aerospace research
9. Integrated systems in aircrafts and missiles
10. Industrial and Military applications